

EMERGENT PHASE 3 - TECHNICAL SPECIFICATION

Date: February 17, 2026
Version: 3.1 Final
Backend Endpoints: 14 Mobile APIs
Platform: React Native + Expo (Pure JavaScript)

1. ENDPOINT-TO-SCREEN MAPPING

Authentication (3 endpoints)

#	Endpoint	Method	Screen	Purpose
1	/auth/login	POST	app/(auth)/login.js	Login with phone + password
2	/auth/me	GET	App startup + profile.js	Get current user
3	/auth/logout	POST	profile.js	Logout user

Chat (3 endpoints)

#	Endpoint	Method	Screen	Purpose
4	/chatbot/message	POST	app/(main)/(tabs)/chat.js	Send message (creates issues)
5	/chatbot/history	GET	chat.js	Get chat history
6	/images/upload	POST	chat.js	Upload temp image

Issues (2 endpoints)

#	Endpoint	Method	Screen	Purpose
7	/issues	GET	issues.js, not-fixed.js, fixed.js	List issues
8	/issues/{id}	GET	issue-detail.js, *-detail.js	Issue details

Dashboard (6 endpoints)

#	Endpoint	Method	Screen	Purpose
9	/dashboard/manager	GET	dashboard/index.js	Manager cards
10	/dashboard/supervisor	GET	dashboard/index.js	Supervisor cards
11	/dashboard/solver	GET	dashboard/index.js	Solver cards
12	/dashboard/issues/analytics	GET	Issues analytics tab	Issues charts
13	/dashboard/complaints/analytics	GET	Complaints tab	Complaints charts
14	/dashboard/history/analytics	GET	History tab	History charts

Ignored: Endpoints 15-17 (webhooks, health - backend only)

2. FILE STRUCTURE

New Files (13 files)

```

src/config/
├── environment.js          # API_BASE_URL configuration
└── api.js                 # Endpoint constants

src/services/
├── apiClient.js           # Axios + token interceptor
├── auth.js                # Endpoints 1-3
├── chat.js                # Endpoints 4-5
├── upload.js              # Endpoint 6
├── issues.js              # Endpoints 7-8
└── dashboard.js          # Endpoints 9-14

src/components/common/
├── Shimmer.js             # Shimmer loading animation
└── AnimatedButton.js      # Animated button (replaces Button.js)

src/components/skeletons/
├── IssueCardSkeleton.js
├── DashboardCardSkeleton.js
└── ChatMessageSkeleton.js

src/store/slices/
└── settings.js           # Haptics settings

```

Modified Files (11 files)

Screens:

```

├── app/(auth)/login.js
├── app/(main)/profile.js
├── app/(main)/(tabs)/chat.js
├── app/(main)/(tabs)/dashboard/index.js
├── app/(main)/(tabs)/dashboard/issues.js
└── app/(main)/(tabs)/dashboard/issue-detail.js

```

Redux:

```

├── src/store/index.js
├── src/store/slices/authSlice.js
├── src/store/slices/chatSlice.js
├── src/store/slices/issuesSlice.js
└── src/store/slices/dashboardSlice.js

```

3. DATA MAPPING

3.1 Authentication

POST /auth/login

Request:

```

{
  "phone": "9123456789",
  "password": "SecurePass@123"
}

```

Response:

```
{
  "success": true,
  "access_token": "eyJhbGciOi...",
  "refresh_token": "refresh_...",
  "expires_in": 3600,
  "user": {
    "id": 1,
    "name": "Priya Sharma",
    "phone": "9123456789",
    "email": "priya@company.com",
    "role": "supervisor",
    "site_ids": [1, 5, 10],
    "sites": [
      {"site_id": 1, "name": "Veperiy Site", "location": "Veperiy, Chennai"}
    ]
  }
}
```

Storage:

```
AsyncStorage:
  @access_token
  @refresh_token
  @user (JSON.stringify)
```

```
Redux:
  auth.user = user
  auth.isAuthenticated = true
```

GET /auth/me**Request:**

Headers: Authorization: Bearer {token}

Response:

```
{
  "id": 1,
  "name": "Priya Sharma",
  "phone": "9123456789",
  "email": "priya@company.com",
  "role": "supervisor",
  "site_ids": [1, 5, 10],
```

```
"sites": [...]  
}
```

Usage: App startup verification, profile refresh

POST /auth/logout

Request:

Headers: Authorization: Bearer {token}
Body: {}

Response:

```
{  
  "success": true,  
  "message": "Logged out successfully"  
}
```

Actions:

1. Call /auth/logout
 2. Clear AsyncStorage keys
 3. Reset Redux
 4. Navigate to login
-

3.2 Chat Flow

Step 1: POST /images/upload

Request:

FormData:

- image: {uri, type: 'image/jpeg', name: 'photo.jpg'}

Headers:

- Authorization: Bearer {token}
- Content-Type: multipart/form-data

Response:

```
{  
  "success": true,  
  "temp_url": "imagekit://temp-bucket/user1_1708160400.jpg"  
}
```

Step 2: POST /chatbot/message

Request (Create Issue):

```
{
  "message": "Pipe broken at Vepery site, urgent need plumber",
  "attachments": [
    {
      "type": "image",
      "url": "imagekit://temp-bucket/user1_1708160400.jpg"
    }
  ],
  "context": {
    "latitude": 13.083944,
    "longitude": 80.270014
  }
}
```

Response:

```
{
  "reply_text": "✅ Issue created successfully!"

  ID Issue #42
  📍 Site: Vepery Site
  🛠 Assigned to: Ramesh K (Plumber)
  📞 Auto-call initiated at 12:05 PM
  ⌚ Deadline: Today 6:00 PM
  📷 Before image uploaded",
  "intent_detected": "create_issue",
  "data": {
    "issue_id": 42,
    "title": "Pipe broken",
    "site_name": "Vepery Site",
    "assignment": {
      "solver_name": "Ramesh K",
      "solver_phone": "9876543210"
    },
    "before_image": {
      "url": "imagekit://bucket/issues/42-before.jpg"
    }
  },
  "chat_id": 100
}
```

UI Rendering:

Display `reply_text`

Show `issue_id` as link → Navigate to `issue-detail.js?id=42`

Show `before_image` if present

Step 3: GET /chatbot/history

Request:

GET /chatbot/history?limit=50&issue_id=42

Headers: Authorization: Bearer {token}

Response:

```
{
  "messages": [
    {
      "chat_id": 100,
      "user_id": 1,
      "issue_id": 42,
      "role_in_chat": "USER",
      "message": "Pipe broken at Vepery",
      "attachments": ["imagekit://bucket/issues/42-before.jpg"],
      "created_at": "2026-02-17T10:30:00Z"
    },
    {
      "chat_id": 101,
      "role_in_chat": "AI",
      "message": "✅ Issue created!...",
      "attachments": [],
      "created_at": "2026-02-17T10:30:05Z"
    }
  ],
  "pagination": {
    "total_count": 250,
    "limit": 50,
    "has_more": true
  }
}
```

3.3 Issues

GET /issues

Request:

GET /issues?status=ASSIGNED&site_id=1&priority=high&page=1&limit=20
Headers: Authorization: Bearer {token}

Response:

```
{
  "issues": [
    {
      "issue_id": 42,
      "title": "Pipe broken",
      "site_name": "Vepery Site",
      "priority": "high",
      "status": "IN_PROGRESS",
      "deadline_at": "2026-02-17T18:00:00Z",
      "raised_by": {"name": "Priya Sharma"},
      "assigned_to": {"name": "Ramesh K", "skill_type": "plumber"}
    }
  ]
}
```

IssueCard Rendering:

#42 - Pipe broken

 Vepery Site

 IN_PROGRESS

 Due: Today 6:00 PM

 Ramesh K

GET /issues/{id}

Request:

GET /issues/42
Headers: Authorization: Bearer {token}

Response (7 nested sections):

```
{
  "issue_id": 42,
  "title": "Pipe broken",
  "priority": "high",
  "status": "RESOLVED_PENDING_REVIEW",
  "deadline_at": "2026-02-17T18:00:00Z",

  "site": {
    "site_id": 1,
```

```
  "name": "Vepery Site",
  "location": "Vepery, Chennai"
},

"raised_by": {
  "user_id": 1,
  "name": "Priya Sharma",
  "phone": "9123456789",
  "raised_at": "2026-02-17T11:00:00Z"
},

"assignment": {
  "assigned_to": {
    "name": "Ramesh K",
    "phone": "9876543210",
    "skill_type": "plumber"
  },
  "assigned_at": "2026-02-17T11:05:00Z"
},

"images": [
  {
    "type": "BEFORE",
    "url": "imagekit://bucket/issues/42-before.jpg",
    "uploaded_by": {"name": "Priya Sharma"},
    "uploaded_at": "2026-02-17T11:00:00Z"
  },
  {
    "type": "AFTER",
    "url": "imagekit://bucket/issues/42-after.jpg",
    "ai_flag": "OK",
    "ai_confidence": 0.92
  }
],

"call_logs": [
  {
    "attempt_number": 1,
    "status": "MISSED",
    "initiated_at": "2026-02-17T11:05:00Z"
  },
  {
    "attempt_number": 3,
    "status": "ANSWERED",
```



```

        "duration_seconds": 120
    }
],

"complaints": [
    {
        "complaint_id": 8,
        "complaint_details": "Leak still visible",
        "complaint_image_url": "imagekit://...",
        "raised_by": {"name": "Priya Sharma"}
    }
],

"history_preview": [
    {
        "action": "Status changed: IN_PROGRESS → RESOLVED",
        "user": "Ramesh K",
        "timestamp": "2026-02-17T17:05:00Z"
    }
]
}

```

UI Sections:

1. Header (issue_id, title, site, priority, status, deadline)
 2. Raised by (name, phone, time)
 3. Assigned to (name, phone, skill)
 4. Images (before/after with AI flags)
 5. Call logs (attempts list)
 6. Complaints (list)
 7. History (timeline)
-

3.4 Dashboard

GET /dashboard/{role}

Request (Supervisor):

```

GET /dashboard/supervisor
Headers: Authorization: Bearer {token}

```

Response:

```

{
    "role": "supervisor",
    "user": {
        "id": 1,
        "name": "Priya Sharma",

```

```
"my_sites": [
  {"site_id": 1, "name": "Vepery Site"}
],

"issues_card": {
  "total_active": 12,
  "my_issues_raised": 25,
  "pending_my_verification": 3,
  "completed_today": 2,
  "high_priority": 4
},

"complaints_card": {
  "total_pending": 2,
  "raised_by_me": 5,
  "resolved_today": 1
},

"history_card": {
  "total_events_today": 8,
  "my_issues_events": 50
}
}
```

UI (3 Cards):

ISSUES
Total Active: 12 Raised by Me: 25 Pending Review: 3 High Priority: 4

COMPLAINTS
Pending: 2 Raised by Me: 5 Resolved Today: 1

HISTORY
Events Today: 8 My Issues: 50 events

Role Differences:

- Manager: All sites data
 - Supervisor: site_ids sites only
 - Solver: My assignments only
-

GET /dashboard/issues/analytics

Request:

GET /dashboard/issues/analytics?date_range=7d

Headers: Authorization: Bearer {token}

Response:

```
{
  "line_chart": {
    "title": "Issues Over Time (Last 7 Days)",
    "labels": ["Feb 11", "Feb 12", "Feb 13", "Feb 14", "Feb 15", "Feb 16",
    "datasets": [
      {
        "label": "New Issues",
        "data": [5, 8, 6, 7, 9, 5, 8],
        "color": "#3b82f6"
      },
      {
        "label": "Completed",
        "data": [4, 6, 5, 6, 7, 4, 5],
        "color": "#10b981"
      }
    ]
  },

  "bar_chart_by_site": {
    "title": "Issues by Site",
    "labels": ["Vepery", "Taramani", "Guindy"],
    "datasets": [
      {
        "label": "Open",
        "data": [3, 2, 4],
        "color": "#ef4444"
      }
    ]
  },

  "pie_chart_by_status": {
    "title": "Issues by Status",
```

```
    "labels": ["Completed", "In Progress", "Assigned", "Open"],
    "data": [25, 12, 10, 8],
    "colors": ["#10b981", "#f59e0b", "#3b82f6", "#6b7280"]
  },

  "summary_stats": {
    "total_issues": 57,
    "avg_resolution_time_hours": 6.5,
    "completion_rate_percent": 78.5
  }
}
```

Similar structure for:

- /dashboard/complaints/analytics
 - /dashboard/history/analytics
-

4. VISUAL POLISH

4.1 Shimmer Loading

Specifications:

Duration: 1500ms infinite
Gradient: 3 colors sliding left → right
Light theme: #E0E0E0 → #F5F5F5 → #E0E0E0
Dark theme: #2A2A2A → #3A3A3A → #2A2A2A
Animation: translateX(-100% → 100%)

When to Show:

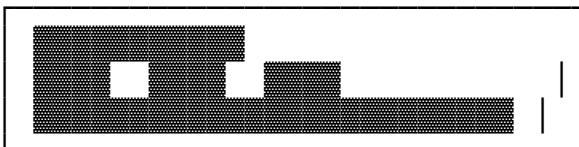
- Initial API load only (not refresh)
- No cached data available

Skeleton Components:

IssueCardSkeleton:



DashboardCardSkeleton:



4.2 Animations

Button Press:

Scale: 1.0 → 0.95 (50ms ease-out)
Opacity: 1.0 → 0.8
Haptic: Light impact

Release:

Scale: 0.95 → 1.0 (150ms spring)
Opacity: 0.8 → 1.0

Success (API OK):

Checkmark scale: 0 → 1.2 → 1.0 (spring)
Background pulse: 0 → 1.1 → 1.0
Duration: 300ms + 2s display
Haptic: Success notification
Auto-hide: 2 seconds

Error (API Fail):

Shake: -10, 10, -8, 8, -5, 5, 0 (50ms each)
Haptic: Error notification (double pulse)
Border: Red pulse

Screen Entrance:

FadeIn: opacity 0→1 + translateY 20→0 (300ms)
SlideIn: translateX -100→0 (400ms spring) for modals

4.3 Haptic Feedback

Patterns:

Action	Pattern	Trigger
Button tap	Light impact	All button presses
List item tap	Medium impact	Issue/complaint tap
Success	Success notification	API success
Error	Error notification (double)	API error
Filter change	Selection	Filter applied

Settings:

Profile screen toggle:

- Switch: "Haptic Feedback"
- Storage: AsyncStorage @haptics_enabled
- Redux: settings.hapticsEnabled
- Default: true

5. PERFORMANCE

5.1 FlashList

Implementation:

```
import { FlashList } from "@shopify/flash-list";

<FlashList
  data={issues}
  renderItem={({ item }) => <IssueCard issue={item} />}
  estimatedItemSize={120}
  keyExtractor={ (item) => item.issue_id.toString() }
/>
```

Estimated Sizes:

- IssueCard: 120px
 - DashboardCard: 200px
 - ChatMessage: 80px
 - ComplaintCard: 100px
-

5.2 Targets

Metric	Target
Button response	< 50ms
Screen transition	< 350ms
API timeout	30s
Upload timeout	60s
Memory	< 150MB
Frame rate	60fps

6. ERROR HANDLING

6.1 Scenarios

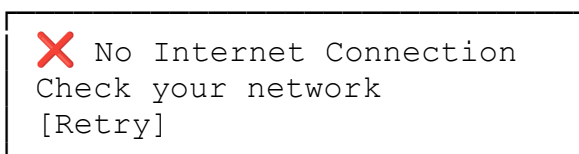
401 Unauthorized:

Action:

1. Clear AsyncStorage (@access_token, @refresh_token, @user)
2. Reset Redux auth state
3. Navigate to login
4. Toast: "Session expired. Please login again."

Network Error:

UI:



Action:

1. Show error state
2. Keep cached data visible
3. Retry button calls API again

API Error (400/500):

UI:

 Something went wrong
{error.message}
[Retry]

Action:

1. Parse error.response.data.message
2. Show ErrorShake animation
3. Trigger error haptic

Timeout:

UI:

 Request Timed Out
Server taking too long
[Retry]

Action:

1. Cancel request
2. Show timeout message
3. Retry with increased timeout

6.2 Offline Mode

Strategy:

1. Check network on app startup
2. If offline:
 - Show banner: "Offline mode"
 - Load cached data from AsyncStorage
 - Disable create/update operations
 - Show read-only views
3. When online:
 - Hide banner
 - Refresh all data

Cached Keys:

@cached_issues
@cached_dashboard
@cached_chat

7. DEPLOYMENT CHECKLIST

7.1 Backend Configuration

- ✓ API_BASE_URL set:
Dev: http://192.168.1.100:3000
Prod: https://api.emergent.com
- ✓ All 14 endpoints tested with Postman

- ✓ Test credentials verified:
Manager: [phone]/[password]
Supervisor: [phone]/[password]
Solver: [phone]/[password]
 - ✓ Image upload works (max 5MB)
 - ✓ JWT expiry: 1 hour
-

7.2 Redux Store

- ✓ All slices use services (not mocks)
 - ✓ AsyncStorage keys:
@access_token
@refresh_token
@user
@haptics_enabled
 - ✓ Token refresh on 401
 - ✓ Persisted state works after restart
-

7.3 Visual Polish

- ✓ Shimmer on all initial loads
 - ✓ Skeleton components match real layout
 - ✓ AnimatedButton on all CTAs
 - ✓ Success/Error animations work
 - ✓ Haptics toggle in profile
 - ✓ All animations at 60fps
-

7.4 Performance

- ✓ FlashList on all lists
 - ✓ estimatedItemSize set correctly
 - ✓ No scroll jank (test on low-end device)
 - ✓ Button response < 50ms
 - ✓ Screen transitions < 350ms
 - ✓ Memory < 150MB (100+ issues loaded)
 - ✓ Image placeholders while loading
-

7.5 Testing

- ✓ Login works for all 3 roles
 - ✓ Dashboard shows role-specific data
 - ✓ Issues list filters work
 - ✓ Issue detail shows all 7 sections
 - ✓ Chat sends messages + images
 - ✓ Image upload → message flow works
 - ✓ Pull-to-refresh works
 - ✓ Logout clears storage
 - ✓ 401 auto-logout works
 - ✓ Network error handling works
 - ✓ Offline mode shows cached data
-

7.6 Platform Testing

- ✓ iOS physical device (iPhone 12+, iOS 15+)
 - Haptics work
 - Camera/gallery work
 - ✓ Android physical device (Android 10+, API 29+)
 - Haptics work
 - Camera/gallery work
 - Back button handling
-

8. DEPENDENCIES

Install Commands

```
# Phase 3 dependencies
npx expo install react-native-reanimated
npx expo install expo-linear-gradient
npx expo install expo-haptics
npm install @shopify/flash-list
npm install axios
```

Already Installed (Phase 1+2)

```
expo-router
@react-navigation/native
redux
```

@reduxjs/toolkit
react-redux
@react-native-async-storage/async-storage
expo-image-picker
expo-camera
expo-location

APPENDIX: QUICK REFERENCE

Endpoints

Auth: POST /auth/login, GET /auth/me, POST /auth/logout
Chat: POST /chatbot/message, GET /chatbot/history, POST /images/upload
Issues: GET /issues, GET /issues/{id}
Dashboard: GET /dashboard/{role}, GET /dashboard/{type}/analytics

AsyncStorage Keys

@access_token	JWT token
@refresh_token	Refresh token
@user	User object (JSON)
@haptics_enabled	"true" "false"
@cached_issues	Cached issues
@cached_dashboard	Cached dashboard
@cached_chat	Cached messages

Performance Targets

Button: < 50ms
Transition: < 350ms
API: 30s timeout
Upload: 60s timeout
Memory: < 150MB
FPS: 60fps

END OF TECHNICAL SPECIFICATION